Admissibility: What a Manipulation Benchmark Score Certifies

Anonymous submission

Abstract

A manipulation success rate can be made of *location* instead of *looking*. We make that precise, measure it, and then report an experiment that refutes our own first reading of it.

Define the *placement radius* R of an object as the radius of the smallest disc containing every position a benchmark ships for it. A single task-indexed constant serves that object if and only if $R \leq \delta$, the tolerance below which the embodiment cannot tell two placements apart; when that holds for every object of every task, no function of episode outcomes distinguishes a lookup table from a policy that perceives accurately.

Measured across all four LIBERO suites [7] from the shipped initial states alone — no policy, no simulator, no training — one suite is the outlier on the object its tasks are *named* by. All ten LIBERO-Object tasks pin that object to $R \leq 1.17\,\text{cm}$; among the other thirty tasks only two do, and both are fixtures bolted to the scene. Six of LIBERO-Object's ten targets take just two `float64` values, one unit in the last place apart, across all fifty shipped states. No suite pins the shared container, LIBERO-Object included.

We then run the lookup policy the proposition describes, on all ten tasks. It scores 16/100 against our released policy's 8/100 — and **that comparison supports nothing**, which is the paper's second contribution.

All twelve discordant trials fall in two tasks and ten of them in one, so the trial-level test we first reported ($p = 0.039$) is inflated by a design effect of 8.4; at the task level it is $p = 1.0$. Worse for us, tasks handed constants identical to within a millimetre score anywhere from 0/10 to 10/10, and success correlates *positively* with placement radius ($r = +0.52$) where the proposition requires the opposite. The goal supply is not what produced the successes.

So: the benchmark measurement stands on its own and is cheap to run before any model is trained; the behavioural claim we built on it does not, and we say which experiment would settle it. Along the way, two results about certification that do survive: two heads whose attribution profiles agree to 0.73 cm score 0.000 and 0.700 deployed (two *earlier* heads, not the released one), and holding weights, seeds and every package version fixed while changing only the renderer leaves the success *rate* identical and flips 40% of the individual episodes.

	probes from	cannot say
Benchmark degeneracy, outside robotics		
Torralba & Efros [13]	dataset identity	—
HANS [9], artifacts [4]	hypothesis only	where in a stack
ImageNet-v2 [11]	a second test set	—
Shortcuts [3]	general statement	—
Perturbation, in robotics		
LIBERO-PRO [14]	re-running with perturbed scenes	how much was gameable *a priori*
LIBERO-Plus [2]	seven axes, ten VLAs	which stage reads the shortcut
robomimic [8], COLOSSEUM [10]	re-collection, perturbation	what the shipped files already permit
This paper	**shipped files, before training**	—

Table 1: Every prior entry needs a trained model and a run to say anything. Placement diameter is computed from the benchmark's own initial-state files with no policy, no simulator and no training, which is what lets it bound what a score *could* certify rather than report what one model did.

1 The question

A policy scores 0.700 on a manipulation benchmark — 35 successes in 50 trials on one LIBERO-Object task, and it is our own policy. What has been certified?

The intended reading is a capability claim: the policy perceives the named object, locates it, and manipulates it. The reading this paper is about is narrower and much cheaper: the policy has learned *where the object always is*. Both readings produce the same number. Distinguishing them is not a matter of looking harder at the policy — two policies can be indistinguishable to a probe and opposite in the world, and we exhibit such a pair — but of first measuring the benchmark.

The framing is not new either. While this work was in revision, Jiang et al. [5] defined *shortcut solvability* — a benchmark score is evidence of capability only if every policy reaching it has the capability — audited five manipulation benchmarks including all of LIBERO, and showed a 0.09B probe with no language encoder scoring at or near reported state of the art on LIBERO-Object. That is our Corollary 1 in content and a stronger empirical demonstration of it than ours. Our *blind* arm is

likewise the manipulation instance of the unimodal ablation that Thomason et al. [12] established in embodied AI. We claim no novelty for either move.

What remains ours is the *a priori* half. Their diagnostic requires training a probe and running it; R is a property of the shipped files, so it bounds what a score could certify before a GPU is switched on, and it says *which* object in a task carries the degeneracy. Their result is the symptom; the radius is a candidate explanation for it that costs a script. Table 1 places the rest. Measuring a benchmark's degeneracy before crediting a model is standard outside robotics, and our *blind* arm is the manipulation instance of the hypothesis-only baseline; we claim no novelty for the move. That VLAs ignore language is likewise not our finding.

What that literature lacks, and what manipulation makes possible, is a metric *computable from a benchmark's shipped files before any model is trained*. The prior here is not a class frequency but a physical coordinate, and a physical coordinate can be measured exactly.

What those results leave open is the part a practitioner needs. They show *that* scores collapse under perturbation. They do not say how much a given benchmark could have been gamed in principle, which stage of a stack is reading the shortcut, whether the shortcut can be removed, or what evidence would count as showing that it had been. This paper answers those four questions for one suite and one stack, in that order, and the first answer is the one that gives the others their meaning.

Contributions. A measurement of a benchmark that runs before any model is trained (§2); a direct measurement of the tolerance assumption it rests on (§5); the withdrawal of our own behavioural reading of it, with the analysis that forced the withdrawal (§5); a four-layer mechanism audit, two of whose layers sit outside the model (§6); and a falsification record (§10).

What is regenerable, and what is not. `scripts/verify_paper_numbers.py` recomputes every number in §2 and §5 *from the per-trial outcomes and the shipped initial states* — including every hypothesis test — and fails on disagreement. An earlier version of that script checked p-values by reading them back out of the JSON that wrote them, which is a consistency check between two copies of a number and not a re-derivation; a reviewer caught it and it is fixed. The script names the claims it does *not* cover, and one of them matters: the historical $0.700 = 35/50$ cell that opens this paper was measured before we shipped per-trial logs, and its trial record is not in the repository.

Scope, stated once and up front. All results are simulation. Every confirmatory cell is one task and one object unless stated. Cell sizes are given with every number and no cell is quoted without its interval. Our external validation attempt is reported as a *negative* (§9): we could not reproduce a public checkpoint's published performance on our build, and we do not report probe results on a harness whose baseline we cannot reproduce.

2 Admissibility

We want a property of a *benchmark*, computable before any model exists, that bounds how much of a score a lookup table could have taken.

2.1 Definitions

Fix a task t that ships a finite set S_t of initial states. A task requires the policy to localise one or more objects; write O_t for that set, which the benchmark declares (LIBERO ships it as `obj_of_interest`), and $x_t^o(s) \in \mathbb{R}^2$ for object o's table position in state s. All norms are Euclidean unless subscripted; §B reports the ℓ_∞ variant, which changes one suite's verdict and not the comparison.

Definition 1 (Placement radius)**.** *The* placement radius *of object o in task t is the radius of the smallest disc containing every shipped position,*

$$R_t(o) \;=\; \min_{c \in \mathbb{R}^2} \; \max_{s \in S_t} \|x_t^o(s) - c\|.$$

R is the quantity the next proposition needs, and it is the quantity because a single constant serves o *if and only if* $R_t(o) \le \delta$ — the minimising c *is* that constant.

An earlier version of this paper used the *diameter* $D_t(o) = \max_{s,s'} \|x_t^o(s) - x_t^o(s')\|$. That criterion is sound where it fires ($D \le \delta \Rightarrow R \le \delta$, since $R \le D$) but it is not tight: $R \ge D/2$ always, and Jung's theorem gives $R \le D/\sqrt{3}$ in the plane. So $D > \delta$ licenses *nothing*, and we had used it to claim that no task outside LIBERO-Object was admissible.

Recomputed on R at that version's $\delta = 2.5\,\text{cm}$, 27 of 38 tasks are admissible, not 10. **The separation we reported was in part an artifact of a loose bound.** What survives it is stated below, on the correct quantity and at a tolerance we measure rather than assert.

Definition 2 (Order-k lookup-admissibility)**.** *Object o of task t is* lookup-admissible at δ *if $R_t(o) \le \delta$. A suite is* order-1 lookup-admissible *at δ if every object of every task is.*

Which objects, and why it matters. Every LIBERO task requires at least two: the thing to be moved and the place to put it. We report them separately, because the split is the finding rather than a technicality. Call o *identity-bearing* if it is what individuates the task from its siblings — the grocery in "pick up the *alphabet soup* and place it in the basket" — and *shared* otherwise. LIBERO-Object's ten tasks differ only in the identity-bearing object; all ten share one basket.

An earlier version measured the identity-bearing object alone and called the suite admissible. That was measuring half of each task. Measured over both, *no* LIBERO-Object task is admissible, because the shared basket has $R \in [1.65, 1.98]$ cm and moves every episode.

Assumption A1 (δ-robustness). Supplying the controller a constant c with $\|x_t^o(s) - c\| \le \delta$ leaves the episode outcome unchanged, for every s. A1 is an empirical property of an embodiment, not a definition. §5 measures it directly and finds a radius of about 1.4 cm on this controller — and finds that the radius is not the same number on every task, which is a real limitation of what follows.

Proposition 1 (Lookup sufficiency)**.** *If $R_t(o) \le \delta$ for every object of every task in a suite and A1 holds at δ, then a policy that reads only the task index, emits the corresponding constants, and is otherwise driven by a controller whose behaviour depends on the supplied goal and not on object identity, achieves on that suite whatever that controller achieves given the true positions.*

Proof. Take c_t^o to be the centre of the minimum enclosing disc. By Definition 1, $\|x_t^o(s) - c_t^o\| \le R_t(o) \le \delta$ for every s, so $t \mapsto (c_t^o)_{o \in O_t}$ is a lookup table meeting Definition 2. A1 applies at each state, giving the same outcome as the true positions; averaging over S_t gives the claim. □

The mathematics is trivial and we do not pretend otherwise. All of the content sits in A1, which is why §5 is about measuring A1 rather than about the proposition.

Corollary 1 (No score can separate them)**.** *On a suite satisfying the hypotheses, no function of episode outcomes distinguishes the lookup policy from any policy that supplies the controller a position within δ of the truth on every shipped state.*

Corollary 1 is a statement about a *comparison class*, and the class is narrow: it contains only policies that are accurate everywhere. No policy in this paper is in it, ours least of all. So the corollary bounds what a score *could* certify; it does not license reading any particular pair of measured cells as confirming it. We flag this because an earlier version of this paper made exactly that mistake in both directions.

2.2 Measurement

Extracting a position from a shipped state is not quite mechanical — a naive fixed-column layout is wrong for every suite containing an articulated fixture — so we compile each task's model and read `jnt_qposadr`; Appendix A gives the procedure and the cross-check.

Two LIBERO-Goal tasks name a fixture with no free joint (a cabinet drawer, a stove). An earlier version excluded them as having "no placement to measure". That was the wrong call in the direction that flattered us: a fixture that never moves has $R = 0$ and is the *most* lookup-admissible object there is. They are counted below and identified.

	identity-bearing R (cm)		shared R	admissible
suite	mean	range	mean	at 1.4 cm
Object	**0.30**	0.00–**1.17**	1.83	**10/10**
Spatial	2.15	1.49–4.49	1.83	0/10
Goal	1.48	0.00–2.48	0.42	2/10†
Long	3.10	2.85–3.28	1.84	0/10

Table 2: Placement radius of every LIBERO task, computed from the shipped initial states alone — no policy, no simulator, no training. The last column counts tasks whose identity-bearing object a single constant serves at the tolerance we measure for this controller (§5). **LIBERO-Object pins all ten of the objects its tasks are individuated by; elsewhere two of thirty**, and †both of those are the fixture tasks — a drawer and a stove — which have $R = 0$ because they are bolted down, not because a randomisable placement was frozen. *No* suite is admissible on the shared container, LIBERO-Object included. Artifact: `results/admissibility.json`.

Table 2 is the measurement, and it says one thing precisely.

> **On the object that individuates its tasks, LIBERO-Object is lookup-admissible on all ten. Among the other thirty tasks, two are — and both are fixtures that never move at all.**

The threshold is not doing the work. LIBERO-Object's largest identity radius is 1.171 cm; the smallest among the other suites' twenty-eight *movable* targets is 1.492 cm. Every δ in that gap returns the same verdict, and the radius we measure for this controller (≈ 1.41 cm, §5) falls inside it. That is the one coincidence in this paper that went our way, and we note it is a coincidence: had the measured radius come out at 2 cm, LIBERO-Goal would be 10/10 too.

How pinned is pinned. In 6 of Object's 10 tasks the target's start position takes exactly *two* `float64` values, one unit in the last place apart on a single axis — constant to the precision the format can express. The other four have $R \in [1.17, 1.19]$ cm.

Across tasks the structure is tighter still: the ten targets occupy two table positions 22.1 cm apart, five tasks each (Figure 1A), so a two-constant table covers the identity-bearing half of the whole suite.

This is a property of the shipped files, not of the task definition. The BDDL for task 0 declares a 5×5 cm sampling region for the alphabet soup. The fifty shipped states place it at one point. So the pinning is an artifact of state generation, not of benchmark design —

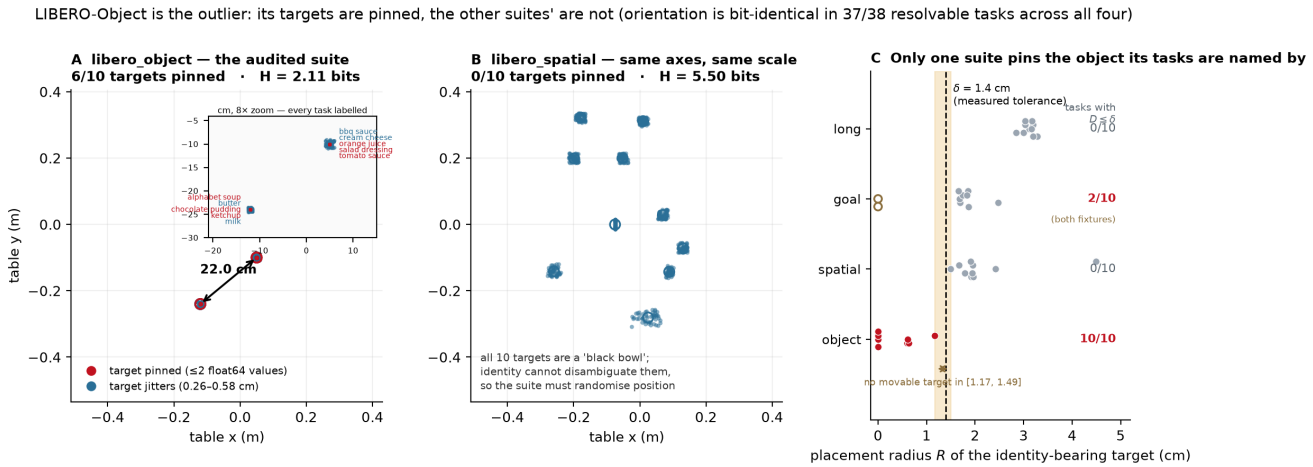

Figure 1: Placement forensics. **A** LIBERO-Object's identity-bearing targets, 10 tasks × 50 shipped states; six collapse to a point and the ten task means occupy two clusters 22.1 cm apart. **B** LIBERO-Spatial on identical axes: every target scatters, and it must, because all ten of its targets are a "black bowl" and identity cannot disambiguate them. **C** placement radius per task against the tolerance measured for this controller. Generated by `paper/render_fig1_placement_forensics.py`; SHA-256 digests of every init-state array are in `results/suite_forensics_columns.json`.

which makes it cheap to fix, and is the strongest argument in this paper for re-sampling rather than redesigning.

What we do not claim. That a low radius *causes* a policy to memorise; that any published policy exploits it; or that admissibility on the identity-bearing object is sufficient for a constant to solve a task. §5 shows the last of those failing on our own stack, decisively.

3 A glass-box stack

The artifact is an instrument, not a contribution. It is built to be traceable end to end, and its size (30.2 M deployed) is what makes an end-to-end trace feasible rather than a claim about efficiency.

component	params	trained	deployed
YOLO-World-S detector	13.0 M	never	frozen
world model (TRM)	9.97 M	stage A	frozen, inert†
trunk heads	7.0 M	stage A	frozen
goal heads + gates	0.24 M	stage B	frozen
deployed	**30.2 M**	17.2 M trained	all frozen

Table 3: Parameter ledger. There is *no* language model: the open-vocabulary detector's own text tower supplies every text embedding, so one frozen network is the only vision encoder *and* the only text encoder. †Inert in the one cell we have ablated: a persistence baseline reproduces the held-out cell exactly. We state it for that cell only.

Text is parsed into source/target phrases; the detector is prompted with them and returns a source box. A grasp head maps (uv, conf, proprio, box_emb, frame_emb) to a world grasp point — note the absence of any text argument, which is Figure 5's point and §8's architectural evidence. A place head maps the command embedding to a place (x, y), latched once per episode inside `reset()` and never re-read. A non-learned state machine converts goals into servo commands within a ±6 cm probe envelope.

Three words, used throughout.

- **cell** — one evaluation: a fixed (head, flags, seed, n) run to completion, always quoted with its n and its interval.
- **band** — the set of shipped initial states a cell replays. The harness fixes it; it is not sampled.
- **burned** — a band we have looked at and let influence a choice. From that moment it no longer measures generalisation, and no amount of re-running un-burns it.

Protocol. LIBERO at commit `8f1084e3`, MuJoCo 2.3.7, robosuite 1.4.1, torch 2.8.0, ultralytics 8.4.115, numpy 2.2.6, Python 3.10, `MUJOCO_GL=osmesa` — §8 explains why that last entry is not boilerplate.

Which states compose a cell is derivable from its command line, not logged. The harness sets `trial_seed` = `seed` · 1,000,003 + `trial` and indexes state `trial_seed` mod 50; since 1,000,003 ≡ 3 (mod 50), trial i at seed s replays state $(3s + i)$ mod 50:

The answer is not flattering: **the n=50 cells are not disjoint from the dev and held-out bands — they contain them.** Fifty trials over fifty shipped states is the whole set. The untouched band exists to supply one clean cell, and §7 reports it.

cell	seed, n	states
dev	0, 10	0–9
held-out	20, 10	10–19
held-out n=50	20, 50	**0–49**
untouched	40, 30	**20–49**

Table 4: Which of the 50 shipped initial states composes each evaluation cell, derived from the harness's seeding rule rather than logged. Because `trial_seed` = `seed` · 1,000,003 + `trial` and 1,000,003 ≡ 3 (mod 50), trial i at seed s replays state $(3s + i) \bmod 50$. The bolded rows are the ones that matter: an n=50 cell is the entire state set and therefore *contains* the dev and held-out bands rather than being disjoint from them.

4 Executing the proposition

Proposition 1 describes a policy. This section runs it, on the one task where our stack works at all, and §5 runs it on the other nine — where it refutes the reading we had given it.

The vehicle is our own stack (§3): a learned goal head that emits a world grasp point, driving a non-learned pick-and-place controller. That factorisation is what makes the experiment clean — we replace the *goal supply* and change nothing else. Figure 2B lists the five arms and what each may read. Two need a word. **random** draws from the observed extent of the scene's objects, re-drawn per episode — the controller's floor. **reset-oracle** reads the true position *once* at reset and holds it; on a task with $R = 0$ that is numerically a per-task constant, but it still reads the simulator, so it is **not** Proposition 1's policy.

The floor is zero. With an uninformed goal the machinery scores 0/10 $[0.00, 0.28]$: eight discordant pairs, all favouring the learned head, exact McNemar $p = 0.0078$. **The controller cannot do this task without a goal.** That is the control that makes the rest of the section mean anything.

The strictly-blind policy. The **blind** arm takes the task index, opens the shipped initial-state files, and emits two constants — the mean shipped position of the identity-bearing object and of the container. On task 0 those are $(-0.1200, -0.2400, 0.0150)$ and $(0.0023, 0.2605)$.

On this task the grasp constant is not an approximation of the target position; it *is* the target position. All fifty shipped states place the alphabet soup at a single (x, y) — the fifty rows reduce to one — so whatever the blind arm fails at, it cannot be error in where it thinks the object is.

It scores 6/10 $[0.31, 0.83]$, against the released head's 8/10. We report no inference from that pair. At n=10 an exact paired test with two discordant pairs returns $p = 0.50$, which is the largest value the test can return with any discordance at all; a power calculation gives 0.058 against the observed effect. **The cell resolves nothing, and an earlier version of this paper reported it as though it resolved indistinguishability.** It does not, and the corollary does not rescue it: Corollary 1 quantifies over policies accurate everywhere, and an 8/100 policy is not one.

Three honest caveats on the arm itself, none of which we found and all of which survived into a previous draft. The constant is the *mean* over all fifty shipped states, which contains the ten this cell replays — it is fitted, very weakly, on its own test set. The arm reads the object's identity from the live scene's body registry rather than from the task index, so "reads only the task index" overstates it. And it requires a live simulator to start at all.

The *control* claim survives all three — with an anchor set, the goal machine is driven by proprioception and the constant, and the grasp head is never called — but the *blindness* claim as we first wrote it did not.

5 The whole suite, and a refutation of our own reading

Every cell above is one task. Run on all ten, the blind constant scores 16/100 and the released head 8/100. An earlier version of this paper led with that pair, paired the 100 trials, ran an exact McNemar, obtained $p = 0.039$, and concluded that a task-indexed constant significantly beats a trained vision-language policy.

That inference is wrong, and the error is instructive.

The unit of analysis. All twelve discordant pairs live in two of ten tasks, and ten of them in one (Table 5). Within that task the ten "trials" are near-identical replays: blind terminates at steps 210–214 (sd 1.08), the head runs to the 300 cap on all ten. They are one deterministic contrast counted ten times.

At the task level — the unit the design actually randomises over — blind wins one task, loses one, and ties eight: exact sign test $p = 1.0$, exact sign-flip permutation $p = 1.0$, cluster bootstrap on the difference $[-0.06, +0.30]$. The design effect is 8.4, so the effective n is about 12, not 100.

unit	test	result	
trial (n=100)	exact McNemar	10–2, $p = 0.039$	*invalid*
task (n=10)	exact sign	1–1, 8 ties, $p = 1.0$	valid
task (n=10)	permutation	$p = 1.0$	valid
task (n=10)	cluster bootstrap	$[-0.06, +0.30]$	valid

Table 5: The same comparison at two units of analysis. Trials within a task share an object, a scene, a goal supply and a near-deterministic trajectory, so the trial-level test is anti-conservative by a design effect of 8.4. **There is no suite-level difference between the two policies at the unit this design randomises over.** Artifact: `results/suite_inference.json`.

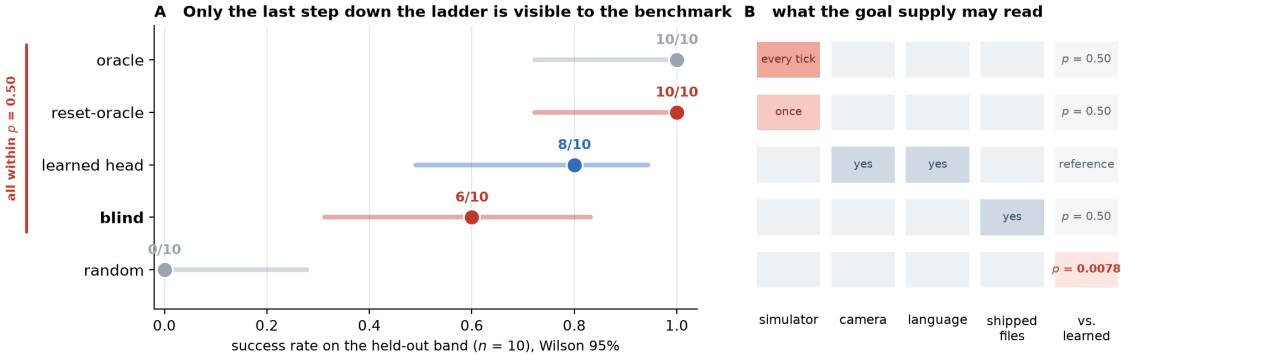

Figure 2: **The goal-supply ladder**, on task 0. Five arms, one controller, one trunk, one protocol, one set of initial states; the only difference is where the goal comes from. **B** is the experiment: descending the ladder removes information. **A** is the result — and the honest reading of it is that at n=10 this cell resolves almost nothing. The bracket marks arms the cell fails to separate, which is a statement about its power, not a finding of equivalence. Artifacts: `results/pod_cells.json`, `results/blind_cells.json`.

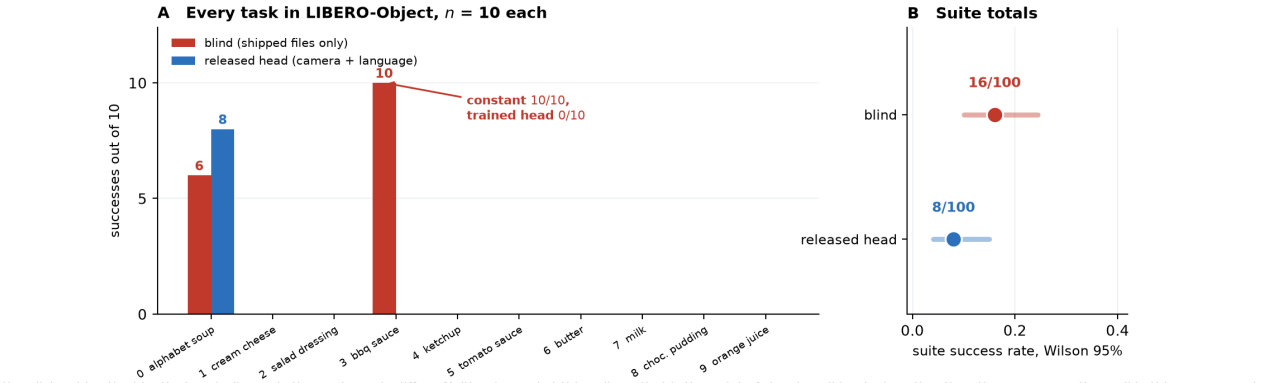

Figure 3: **All ten LIBERO-Object tasks**, n=10 each, every trial recorded; both policies drive the identical controller. Eight of ten tasks are 0–0. The suite totals in **B** are shown with i.i.d. Wilson intervals, which are *too narrow* here by a design effect of 8.4 (§5); they are drawn as reported rather than as believed. Artifacts: `results/suite_cells.json`, `results/suite_inference.json`.

The constant is not what produced the successes. This is the sharper finding, and it comes from the same run.

If the blind arm's score came from the constant being close to the truth, tasks handed the *same* constant would score alike. Table 6 shows they do not. The ten tasks fall into two groups by table position; within each, the constants agree to under a millimetre; and within each, the scores range from 0 to 10.

What varies across those tasks is the object, not the number. So the blind arm's two non-zero cells are evidence that *this controller can grasp two of ten groceries*, and not evidence about lookup. **We withdraw the suite-level claim.**

Where A1 does break, we can see it. One place in this data does behave the way the proposition expects. On task 0 the blind arm reaches the object to within 2 mm on all ten trials — the grasp constant is exact there — and the four failures are the four trials whose *basket* sits furthest from the constant, occupying ranks 6, 8, 9 and 10 of ten (exact Mann-Whitney $p = 0.019$, Figure 4). Task 3 tolerates 1.91 cm and never fails. That gives the ≈ 1.4 cm radius §2 uses, and shows it is task-dependent.

tasks	constants agree to	blind scores
0, 4, 6, 7, 8	0.95 mm	6, 0, 0, 0, 0
1, 2, 3, 5, 9	0.63 mm	0, 0, **10**, 0, 0

Table 6: **The goal supply does not explain the outcome.** Tasks whose constants are identical to within a millimetre score anywhere from 0/10 to 10/10. Tasks given a *bit-exact* constant ($R = 0$) score 6/60; tasks given the loosest constants ($R > 1$ cm) score 10/40. The correlation between placement radius and success is $r = +0.52$, where Proposition 1 requires it to be non-positive. Artifact: `results/constant_attribution.json`.

Two caveats we owe a reader, because this is the num-

ber §2 leans on. The hypothesis was formed *after* a pre-registered one failed, on the same ten points. And the covariate is confounded with trial order: the outcome is exactly $\mathbf{1}\{\text{trial} \leq 5\}$, trial i replays state $10 + i$, and displacement correlates with state index at $\rho = 0.65$, so trial index alone predicts better ($p = 0.0095$). At n=10 the design cannot separate them.

What would settle it. Displace the constant by r and sweep r: if the score is flat in r, the arm never used the number. We implemented it (`--goal-anchor-jitter-cm`) and the compute window closed before it ran. Until it does, the strongest statement the suite supports is the negative one above.

What survives. Two things, and we think they are the more useful two. First, on task 3 a constant scores 10/10 where the trained head scores 0/10 — a single task-level event, so we quote no p for it, but a striking one: the head is not merely worse there, it never succeeds.

Second, and independent of any of this, **eight successes in a hundred trials, every one on the single object the head was trained for**. Our artifact is a one-object solution that a suite-level score would have reported as a policy. The number a reader should carry away is not 0.700; it is 0.700 on task 0, with 0.000 next to it nine times.

And it bounds the audit in advance. Whatever §6–§7 bought, it was not a general policy. We put this section before the audit so that no reader reaches the repairs holding a larger idea of the artifact than the artifact supports.

6 The audit: four layers

"The policy memorised the location" is not one hypothesis but four, at four stages — causal confusion [1] spread across a pipeline rather than confined to a network — two of which are not in the model at all — and those two are the ones a black-box perturbation study cannot reach.

Three of the four are described here; the fourth, being the one the instruments found rather than confirmed, gets its own subsection.

Layer 1: the expert's calibration constant. The scripted demonstrator carried a hand-eye offset fitted once, on a scene whose target never moved. That constant therefore *encoded* the pinned placement, and every demonstration inherited it — so the corpus taught the pose before any network saw a pixel. Detected by reading the expert, not the policy. Repaired by re-baking ten episodes with the source object displaced, which breaks the constant's validity and forces the demonstrator to look; the repaired corpus lifts the dev cell from 0/10 to 7/10 on the same episodes.

Layer 2: the selection loop. Our own procedure, not the artifact's: at each round we chose a checkpoint using cells that later rounds reused, so selection pressure accumulated on one band of initial states. This is invisible to any measurement of the model and is the reason §7 runs an untouched band at all. It is *bounded* rather than repaired — one cannot un-look at data — and the bound is that thirty never-inspected states score within noise of the burned ones.

Layer 3: the grasp head's proprioceptive shortcut. Substitution attribution showed the predicted grasp point tracking the end-effector and barely moving under uv substitution: the head had learned to predict where the object is from where the arm is, which on a pinned suite is a sufficient statistic. The same teleported episodes repair it, and the released head's profile inverts — it now moves further on its visual channels (3.43/6.57 cm) than on proprioception (1.93 cm).

6.1 Layer 4: where the language goes

Swapping the instruction while holding the scene and the success predicate fixed collapses the released head from 8/10 to $\mathbf{0/20}$ $[0.00, 0.16]$; on the ten trials that pair with the reference cell, eight discordant pairs all one way, exact McNemar $p = 0.0078$.

Telemetry places the failure *downstream of approach*: the policy still reaches the true object as closely as baseline while the grip-close rate falls from ∼0.67 to ∼0.26. Driving the detection prompts and the place-head embedding from *different* instructions isolates the collapse to one channel, with no interaction.

Combined with the architecture: **object selection, approach and grasping run with no language input at all, and the entire language channel is a single cached** (x, y). Ask this policy for the butter and it finds, approaches and grasps the alphabet soup at baseline rate; it fails only when that one coordinate is wrong.

7 Repairs, and one we withdraw

Three of the four layers admit a repair, and we report what each bought. One claim from the previous version does not survive being run at power, and we retract it here rather than leave it standing.

The untouched band. Both n=50 cells contain the burned states, so we ran the released head on states 20–49, which no selection look ever reached, with the prediction recorded before the run ($[0.50, 0.75]$, falsifiers named in both directions). Result: $\mathbf{20/30 = 0.667}$ $[0.49, 0.81]$, against the published 0.700 (35/50) a difference of -0.033, Newcombe $[-0.24, +0.16]$. That interval assumes independent samples and the samples are *not* independent:

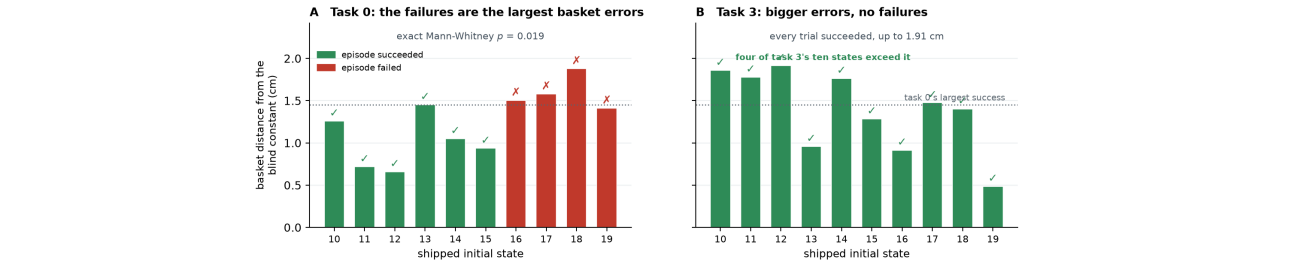

Figure 4: **The one place the constant demonstrably matters. A** On task 0, the trials that fail are those whose basket sits furthest from the single constant the blind arm was handed. **B** On task 3 the same quantity runs higher and every trial succeeds, so the tolerance is not one number across tasks. This is the measurement §2's $\delta \approx 1.4$ cm comes from; see the confound caveats in the text. Artifact: `results/blind_failure_attribution.json`.

layer	where	evidence	repair	verification / status
1. expert calibration constant	scripted expert (non-model)	hand-eye constant encodes the pinned placement	placement teleportation	0/10 → 7/10 dev, paired. **Repaired**
2. selection loop	our own procedure (non-model)	checkpoints chosen on cells later rounds reused	process fix; bands frozen	untouched band 20/30 vs 0.700: no detected inflation. **Bounded, not removed**
3. grasp-head proprio.	`GraspPointHead`	prediction tracks the eef, flat under uv sweeps	same teleported episodes	attribution flips to visual. **Repaired**
4. place-head cmd→location	`PlaceHead`	correct basket for the trained command, 14.5/13.0 cm off for the other two; swap 8/10 → **0/20**, p=0.0078	command coverage	place *point* 14.15 → 0.78 cm **repaired**; swap *behaviour* 0.767 → 0.433, p=0.031 — **halved, not fixed**
(separate) appearance drift	deployment viewpoints	NN-cosine gap on the policy's own trajectory	DAgger on own viewpoints	3/50 → 22/50 vs 3/50 control, $p < 10^{-4}$. **Repaired**

Table 7: Master audit table. Layers 1–4 are the placement-memorisation layers; the last row is a drift case study, *not* one of the four; it appears here only because its repair is easily mistaken for one of theirs. Two rows are deliberately not marked repaired.

the n=50 cell spans states 0–49 and so contains all thirty untouched states.

The interval is therefore conservative in the wrong direction, and we use it only to bound the effect as small. The clean comparison — untouched-30 against the twenty genuinely burned states — is the one we would run next.

We claim exactly this much: not that the burn is zero, but that it is small enough to hide inside ±0.20, which is what n=30 buys.

A repair we withdraw. Command coverage was reported as repairing the instruction-swap behaviour on the strength of 3/10 against a 4/10 baseline at $p = 1.0$. That is an equivalence claim from an underpowered cell, on a band where the head was already mostly failing — and a swap cannot visibly break a cell that is already broken. Re-run on a band where the coverage head scores well, and taken to n=30:

At n=10 this contrast was $p = 0.125$ and we would have called it unresolved; at n=30 it is $p = 0.031$. **The coverage head collapses too.** A swap cannot visibly break a cell that is already mostly broken, which is why the original 4/10 baseline hid the effect.

head	baseline	swapped	paired McNemar
released head	8/10	**0/20**	$p = 0.0078$
coverage	23/30	**13/30**	$p = 0.031$

Table 8: Substituting a different task's instruction, on a band where each head scores well and at a power the original cell did not have. Both heads lose accuracy, so command coverage does not eliminate the instruction dependence it was built to remove; it halves it. Paired McNemar over matched trial indices (same seed, same initial states, same renderer — only the instruction differs), exact two-sided.

What replaces the withdrawn claim is more useful than it was. Coverage does not eliminate the failure, it roughly *halves* it: the released head loses its entire cell (0.800 → 0.000, every discordant pair one way), the coverage head

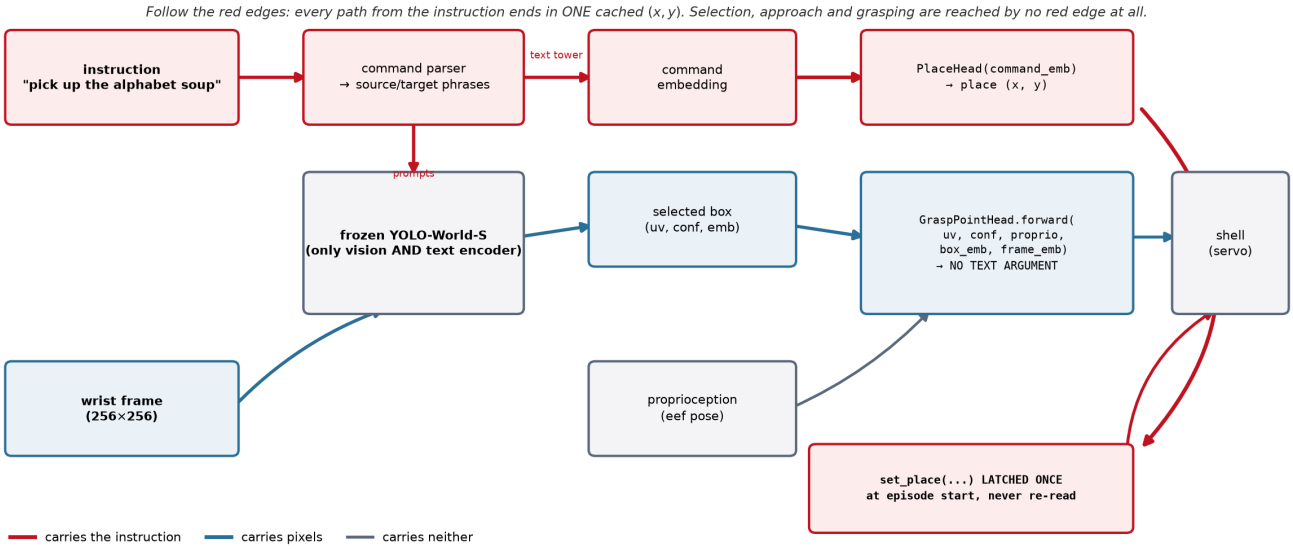

Figure 5: Every edge coloured by what it carries. The architectural half of the argument is a code fact: `GraspPointHead.forward` takes no text argument, so no red edge reaches object selection, approach or grasping, and every red edge terminates in one (x, y) latched once per episode and never re-read.

about a third (0.767 → 0.433). And the repair to the place *point* is not in doubt — 14.15 → 0.78 cm (Figure 6), measured directly rather than inferred from behaviour.

So: **command coverage fixes where the head aims and does not fully fix what it does.** A residual command→location dependence survives training on all three commands — a sharper statement of layer 4 than we had, and one no cell we ran before could have detected.

7.1 Randomisation is a curve, not a point

Our ±4 cm radius was chosen to sit inside the controller's ±6 cm probe envelope — a perturbation calibrated to the system under test. Sweeping it on the same draws separates two things that single number confounded:

displacement	released head	Wilson 95%
none‡	4/10	[0.17, 0.69]
±2 cm	7/10	[0.40, 0.89]
±4 cm‡	4/10	[0.17, 0.69]
±6 cm (envelope edge)	2/10	[0.06, 0.51]
±8 cm (outside)	3/10	[0.11, 0.60]

Table 9: Sweeping the placement-randomisation radius on the same draws. Randomisation robustness is a curve, and at n=10 this curve is not even monotone — the undisplaced cell sits *below* ±2 cm, which cannot be a real benefit of displacement. Reporting any single radius as "passes randomisation" would quote a number this sample does not resolve, and we do not use this table to set any threshold elsewhere in the paper. ‡Per-trial outcomes for the ±2, ±6 and ±8 rows are in `results/pod_cells.json`; the two marked rows were measured in an earlier session and their trial records are not in the repository.

Read conservatively, because n=10 supports no more: clearly worse at ±6–±8 than at ±2, but every interval is ∼0.5 wide and the undisplaced cell sits *below* ±2 cm, which cannot be a real benefit of displacement. That non-monotonicity is the honest measure of what this n resolves. **No single radius should be quoted as "passes randomisation"**, and a benchmark asking for one is asking for a number that does not exist.

8 Certification

If a benchmark score cannot certify grounding, what can? This section reports three results about the certification problem itself, two of them negative.

8.1 Probes alone cannot certify

A substitution probe asks which input channels a head reads: swap one channel between two recorded ticks and measure how far the predicted grasp offset moves. It is annotation-free, cheap, and on-manifold — and that last property is fatal on its own.

One detail in Table 10 decides whether the probe means anything: it profiles the grasp *offset*, not the absolute point. The head emits $\text{eef}_{xy} + \Delta$, so substituting proprioception moves the absolute prediction by the entire arm displacement for *any* head, and profiling that quantity reports ∼24 cm of "proprioception attribution" even for a repaired one.

We made exactly this error when regenerating the measurement. We record it because it is the difference between a probe that answers "does this channel change

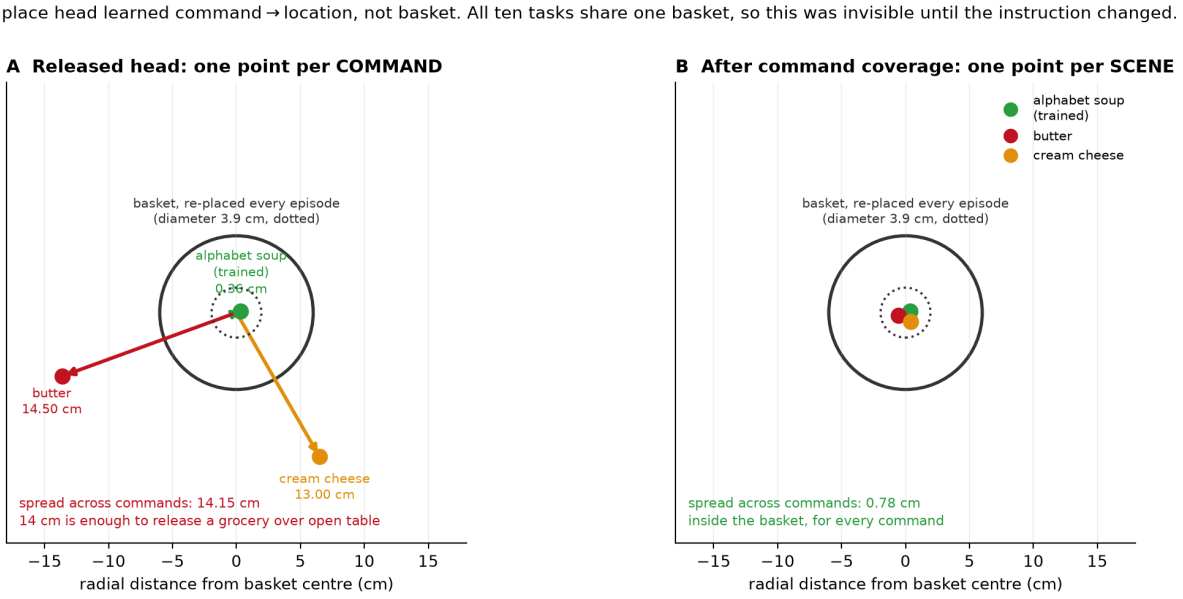

Figure 6: Layer 4 drawn rather than argued. The released head asked for three objects emits three place points, 14.5 and 13.0 cm from the basket. The basket is *not* fixed — it is re-placed every episode, with a placement radius of 1.83 cm averaged over the suite's ten tasks — so a memorised place point is wrong by 14 cm against a target that moves by under 2, and 14 cm is enough to release a grocery over open table. **Only radial distance is measured**: the angles are chosen for legibility and carry no information, which is why no coordinate grid is drawn.

head	uv	proprio	box_emb	frame_emb
v2	0.75	2.85	6.04	6.91
v2.1	0.71	2.13	6.26	7.58
v5 (released)	0.97	1.93	3.43	6.57

Table 10: Substitution-probe attribution, in cm of movement of the predicted grasp *offset*, per input channel. The counterexample is the first two rows: v2 and v2.1 agree to within 0.73 cm on every channel and score 0.000 and 0.700 when deployed — these are heads v2 and v2.1, an earlier pair, and their 0.700 is not the released head's 35/50. A probe evaluated on teacher trajectories cannot, on its own, certify off-manifold behaviour.

where the head thinks the object is" and one that measures the parameterisation.

v2 and v2.1 differ by at most 0.73 **cm on any channel and score** 0.000 **and** 0.700 **deployed**. One caveat we state rather than let a reader discover: the substitution is between a *single* pair of recorded ticks — the two most separated in *uv* out of 329 — with no resampling over pairs and therefore no interval on the 0.73 cm. It is an existence proof that two heads can be probe-identical and behaviourally opposite, not an estimate of how often that happens.

A probe evaluated on teacher trajectories is an on-manifold instrument and cannot, alone, certify off-manifold behaviour. Probe evidence must be paired with behavioural randomisation.

8.2 An instrument only ever shown firing is not yet an instrument

Our cross-instruction detection-agreement probe asks whether two objects' prompt chains select the *same box from the same frame*; a high same-box rate says the grounding stage carries no object identity. Every published number from it was a firing. Running it against every other object in the scene splits cleanly:

counterpart	n	same-box	median dist.
cream cheese, butter, pudding	6	**0.00**	0.817–0.818
tomato sauce	4	1.00	0.00069
bbq sauce, salad dressing	5	0.80	0.0043
ketchup	5	0.60	0.0048

Table 11: The positive control our detection-agreement instrument had never been given. "Same-box" is the fraction of frames on which two objects' prompt chains select the identical box; a high rate means the grounding stage carries no object identity. Run against *every* counterpart rather than only the ones that fire, the instrument separates cleanly — and the split is interpretable: the chains that collide are the cans and bottles, the ones that separate are the boxes. Deployed binding is blind *within* a shape class, which is the regime LIBERO-Object's groceries occupy. Small n (only 4–6 of 60 frames have both chains detecting) makes these demonstrations of discrimination, not calibrated rates.

The instrument registers difference when difference exists. The distances are bimodal — collisions at ∼0.004, separations at ∼0.82, nothing between — and a sweep over $\tau \in \{0.005, 0.01, 0.02, 0.05\}$ leaves every verdict un-

changed. The split is interpretable and sharpens the claim: the chains that collide are the cans and bottles, the ones that separate are the boxes. **Deployed binding is blind *within a shape class***, which is precisely the regime LIBERO-Object's groceries occupy.

Two limits. Only 4–6 of 60 frames had both chains detecting, so these are small-n demonstrations of discrimination, not calibrated rates. And the contrast we originally designated the positive control — the basket — is not evaluable at all: the deployed source-area filter rejects basket-sized boxes by construction, so it compares zero frames. Our first version of the script read same_rate = 0 off that empty comparison and printed "instrument discriminates". A verdict now requires $n \geq 3$.

8.3 Certificates are joint in (head, stack), down to the renderer

We rebuilt the deployment stack on fresh hardware, pinning every version in §3 and verifying both load-bearing checkpoints by digest. The reference cell reproduces: 8/10 $[0.49, 0.94]$ against a recorded 7/10, with 0.700 inside the interval.

Reproducing the *rate* is not reproducing the *experiment*. Holding seed, trial, weights and every package version fixed and changing only the OpenGL backend MuJoCo renders through:

	`osmesa`	`egl`
success rate	8/10	8/10
per-trial outcomes	**4 of 10 disagree** (2 each way)	

Table 12: Changing only the OpenGL backend MuJoCo renders through — same seed, same trials, same weights, same package versions. The aggregate is identical and 40% of the individual episodes are not. Matching a published rate is therefore weak evidence of having reproduced an experiment, and any paired statistic computed across a renderer change is invalid, because pairing assumes the trials are the same trials.

The aggregate is identical and 40% of the episodes are not. Thread count, by contrast, changes nothing we could detect: three `osmesa` runs at 1, 4 and 128 threads agree on all eight logged fields while the wall clock falls from 220 s to 32 s. We state the limit plainly — that comparison ran to 40 steps, a prefix in which the gripper never closes, so it does not probe the contact-rich part of an episode where the renderer difference showed up (`results/thread_determinism.json`). Software and GPU rasterisation do not produce identical pixels, the detector is not invariant to that difference, and in this stack the detector is the only encoder.

Two consequences. Matching a published rate is weak evidence of having reproduced an experiment: the marginal can be right while the sample is different. And *any paired statistic computed across a renderer change is invalid*, because pairing assumes the trials are the same trials.

We must therefore report our own compliance honestly. Every paired test in §5 is within one build, because those cells were produced in one sweep. We cannot say the same for the whole paper: the confound test of §10.1 ran on a different machine entirely — macOS, MuJoCo 3.3.0, robosuite 1.4.0, Python 3.13, recorded in its artifact — and no per-cell build ledger exists for the older cells. Given what this subsection demonstrates, that is a gap in our own protocol and not a footnote. `MUJOCO_GL` belongs in any reported stack; it was not in ours, or in any LIBERO evaluation we have read.

9 Transfer: a negative result

The instruments above are offered as portable, so we tried them on a policy we did not build: the public `openvla-7b-finetuned-libero-object` checkpoint [6], through the same harness, trial-to-state map and success predicate.

The baseline returned 0/1 on a checkpoint published near 0.9. We treated that as a harness defect and found four: image orientation, the gripper convention (their model emits $[0, 1]$ with 1=close, robosuite wants $[-1, +1]$ with -1=close — without the conversion the jaw never closes), success extraction (the environment's `done` flag also fires on horizon), and a missing signal shield. We then swept four image orientations against the 0.9 centre crop their evaluation applies.

The checkpoint still does not approach the target in any configuration we tried. **We therefore report no OpenVLA numbers at all** — not success rates and not distance statistics — because the diagnostic runs were exploratory and were never written to a versioned artifact, and an unvalidated baseline would make our instruments look powerful for the wrong reason.

We cannot separate a remaining defect in our harness from a checkpoint that does not transfer to a differently-built LIBERO, and the reproduction gap we hit is a known community issue rather than a finding of ours. The harness ships (`eval/openvla_eval.py`). Until someone with the matching build runs it, **the portability of these instruments is unestablished**.

10 Negative results and registries

A paper that only reports what worked cannot be checked. This section is the ledger: what we predicted before running, what failed, which instruments we retired, and the three ways our own results fooled us.

Pre-registration. Fourteen predictions are on record with their outcomes; seven failed. Two failures matter here: the coverage head's swap cell did not hold (p=0.031), and the released head was not at floor outside the controller's envelope (3/10).

The blind arm carries *no* registered prediction — it was implemented in one compute window and run in the next — and neither does the attribution analysis that overturned it (§5), which was formed after the registered hypothesis failed. We flag both rather than reconstruct predictions after the fact.

Retired instruments. Quantifying deployed role binding was attempted six times and abandoned six times, each at a calibration gate written down before the instrument was trusted. Deployed binding accuracy is reported as **unmeasured**; the six gates are in the appendix.

10.1 A confound we had missed

Our own analysis of a blocked object was confounded: because the suite's two placement clusters align with which objects we happened to train on, "blocked" and "22 cm from the training position" were perfectly correlated in our design. Tested directly over all ten objects, position predicts the effect weakly (Spearman +0.19, p=0.60) and *teleporting each object 22 cm shows no directional effect* (6/10, exact sign test p=1.0, the least informative value the test returns), while detector groundability predicts it (-0.71, p=0.027). The mechanism is appearance, not position — position is a nuisance term with mean $|\Delta| = 0.0066$, which is not zero either.

This does not contradict §2, and the distinction is worth stating. Those sections answer different questions. Placement is what the suite fails to randomise and therefore what a policy *may exploit*: that is a property of the benchmark, established from its shipped files, and it is what layers 1 and 3 were caught doing. Appearance is what limits *transfer to a new object* once that shortcut is repaired: a property of the frozen detector, and why the released head works on one object and not on nine (§5).

A benchmark can be lookup-admissible and a policy can still fail for an unrelated reason. Both are true here, neither weakens the other, and what *would* have been a contradiction — position explaining both — is what the confound test ruled out.

We also report the caveats on this cell rather than only its conclusion. Both nulls are n=10, which is the size we decline to accept elsewhere (§7); the groundability predictor and the outcome are two statistics of the same detector forward pass, so their correlation is closer to a self-consistency check than to an independent prediction; and $p = 0.027$ across three tests does not survive a Bonferroni correction. The confound is *disconfirmed as a sufficient explanation*, which is what the design can support, and not more.

10.2 Four ways a result fooled us

Recorded because the failure modes generalise and *none* was caught by a conventional check — in all four the instrument was correct and the defect was elsewhere (Table 13). The fourth is this revision's headline.

what we saw	what it was	the general lesson
a perfect null: teleport gaps identical to control for all ten objects, to four decimals	a cached observation — the renderer returned the *pre*-teleport frame	implausible cleanliness is a symptom. The script now compares frames across the intervention and *raises* rather than emitting
a defect count withdrawn as unsupported	the appendix we could not find existed	"I could not find it" is a claim about a search, not about a corpus
7/7 against a published 7/10; we began writing up a failure to reproduce	a censored sample. The complete cell is **8/10**	successes end episodes early and failures run to the horizon, so any interim harvest is biased toward successes. **Verifying the instrument does not verify the sampling**
a constant outscoring our trained policy 16/100 to 8/100, $p = 0.039$	ten replicates of one task counted as ten trials — and a constant that did not cause the successes at all	we checked the numbers and not the *unit*, nor whether the variable we manipulated was the one that moved. **Both errors were found by adversarial review of data we had already published**

Table 13: Four self-inflicted errors and their repairs. The third is the one we would most expect others to hit: every check we ran had passed, and the defect was in *when* we looked. The harvester now reports each cell against its planned n and refuses to report an incomplete one.

11 Discussion

What a score certifies. On a suite whose placement radius sits below the embodiment's tolerance, a success rate certifies that the policy can execute a manipulation *given* a correct target pose — and nothing whatever about how it obtained one. That is a statement about what the score *could* distinguish, and it is exactly as strong as A1.

We are careful here because we already overreached once. Our own cells do not demonstrate that any policy exploited the degeneracy; §5 shows they cannot. What the radius buys is a check that runs before training, and a prediction — that a suite with small radii admits a cheap solution — which Jiang et al. [5] confirm on LIBERO with a trained probe where we did not with a constant.

Shortcuts live in pipelines, not only in models. Two of our four layers are outside the network: a scripted expert's calibration constant and our own checkpoint-selection loop. No amount of perturbing a trained model

from the outside would have found either. This is an argument for glass-box artifacts in benchmark-validity work, and it is the main reason our stack is small.

Certification is a system property. Probes cannot certify alone (two heads, 0.73 cm apart, 0.000 vs 0.700). Behaviour must be paired with randomisation. And the certificate is joint in *(head, stack)* down to a component no requirements file pins: the same weights on the same seeds give the same rate and a different 40% of episodes under a different renderer.

Limitations.

- One simulator, one benchmark suite, one robot, wrist camera only.
- Every confirmatory cell is one task and one object at $n \leq 30$, including the blind arm (n=10). The suite-level claims (§2) are the ones computed from files and do not have this limit.
- The held-out band is partially burned; the disclosure is in §3 and the bound in §7.
- The on-manifold limit of substitution probes is a limit on our own instruments, not only on others'.
- Transfer to an external policy is **unestablished** (§9).
- D and H_δ are defined for suites shipping enumerable initial states, and would need restating for procedurally generated ones.

11.1 Recommendation to benchmark authors

Both quantities are cheap to compute and cheap to report. Table 14 is what we would ask of a suite.

practice	cost	why
publish D and H_δ per task	one script, no training	tells a reader what a score can certify
set a floor on D	re-sample the init files	a suite individuated by identity must randomise placement, or identity is never load-bearing
ship a tier of radii	k extra eval configs	where a policy degrades depends on displacement relative to the controller's envelope (Table 9); one radius is a calibration, not a result

Table 14: Three practices, in increasing order of what they cost a benchmark's authors. LIBERO-Spatial already satisfies the second by necessity — all ten of its targets are the same bowl — and LIBERO-Object could satisfy it by choice.

12 Conclusion

A manipulation benchmark score is not a measurement of capability until the benchmark's admissibility has been measured. The measurement is cheap, it runs before any model is trained, and on LIBERO it says something specific: the one suite whose tasks are individuated by object identity is also the one that pins that object, on all ten tasks, to a radius smaller than the tolerance we measure for our own arm.

The behavioural claim we built on that does not stand, and the way it fell is the more useful half of this paper. We ran the lookup policy, saw it outscore our trained one, tested it at the wrong unit of analysis, and reported a significance that a task-level test does not support. The deeper error was attributional: tasks handed constants identical to within a millimetre score from 0/10 to 10/10, so the successes were never evidence about lookup at all. Both errors were found by adversarial review of a manuscript we already believed, using data we had already published.

What survives is the procedure rather than the artifact: measure the benchmark first, at the unit the design randomises over; check that the variable you manipulated is the one that moved; and report the reading that did not survive being checked, which in this revision is our own.

A Recovering which column owns which object

LIBERO ships each task's initial states as a MuJoCo flattened state $[\,t \mid q \mid \dot{q}\,]$; every shipped $\dot{q}$ is zero, so the columns that vary across states are exactly the position degrees of freedom the suite randomises. Recovering *which* object owns which column requires care. The fixed layout $[\,t \mid 9 \mid N\times 7 \mid \dot{q}\,]$ that works for LIBERO-Object predicts a width of $19 + 13N$ and fails for every suite containing an articulated fixture — LIBERO-Spatial ships width 92 against 5 declared objects. We therefore run two passes: the fixed-layout one, and one that compiles each task's model and reads `jnt_qposadr` with the joint names, mapping every column onto a named body. **The two agree wherever both apply**, and only the second is used for suites the first cannot parse. Artifacts: `results/suite_forensics_columns.json` (digests), `results/suite_forensics_joints.json` (per-object positions).

B The norm

Definition 1 uses the Euclidean norm. A1 is naturally stated in ℓ_∞, because LIBERO randomises placement in an axis-aligned box, and the two differ by up to $\sqrt{2}$ in the plane. Recomputing Table 2 under ℓ_∞ at $\delta = 1.4$ cm leaves LIBERO-Object at 10/10 and moves one LIBERO-Spatial task across the line (1/10 instead of 0/10). The

comparison is unchanged; the exact count in one suite is not. We report the Euclidean version throughout and flag the dependence here rather than leaving a reader to find it.

C Sensitivity of H_δ to the tolerance

The quantisation δ is a modelling choice, so the finding must not depend on it. Recomputing placement entropy across five tolerances spanning two orders of magnitude:

suite	0.5mm	2mm	5mm	1cm	5cm
Object	**2.11**	**0.48**	**0.00**	**0.00**	0.00
Spatial	5.48	4.49	2.04	0.25	0.00
Goal	5.59	5.08	1.53	0.02	0.00
Long	5.63	5.40	4.27	0.37	0.00

Table 15: Mean placement entropy H_δ in bits, swept over an order of magnitude in the tolerance δ. LIBERO-Object is lowest at every tolerance, so the ordering does not depend on the modelling choice — but every suite reaches zero by $\delta = 5\,\text{cm}$, which is a statement about coarse resolution rather than about LIBERO. That degeneracy is why the paper's claim is licensed on the diameter D (Table 2), which has no such parameter. Artifact: `results/placement_entropy.json`.

The ordering is unchanged wherever the measure discriminates at all. Coarsening δ lowers every suite's entropy, as it must — coarser radii merge components — and by $\delta = 5\,\text{cm}$ every suite reaches zero, which is a statement about coarse resolution rather than about LIBERO. This is precisely why the paper's claim is licensed on the diameter D (Table 2), which has no such parameter: at $\delta = 5\,\text{cm}$ every suite would look admissible on H, while on D only LIBERO-Object is, at any tolerance the controller actually has. Artifact: `results/placement_entropy.json`.

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